Membership versus variety, and the partial-solution strata: what changes if the PDE is added to the system before decomposition

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A companion note to *A New Method of Solving PDEs*. The core algorithm *reduces* the PDE P modulo the ansatz A and projects the remainder onto the constants, returning the locus $\mathcal{V}$ of constants for which the ansatz family solves the PDE. A natural-looking alternative is to adjoin P to the ansatz and run a differential decomposition (Rosenfeld–Gröbner or, better, differential Thomas) on the *combined* system $A \cup \{P\}$. This note records why that answers a *different* question, names the discrepancy — the *partial-solution strata* — and exhibits it on a two-line example.

1 Two different loci

Let $A(c) = \{a_1, \ldots, a_k\}$ be the ansatz, a differentially triangular set whose coefficients involve constant parameters $c \in \mathbb{C}^n$, and let P be the PDE. The core method computes the **membership locus**

$$\mathcal{V} \;=\; \{\, c \in \mathbb{C}^n : P \in [A(c)] \,\}, \tag{1}$$

a *universal* condition: *every* function in the ansatz family at c solves the PDE. Operationally, Ritt's full reduction of P modulo A yields a remainder $r = \sum_\alpha m_\alpha\, p_\alpha(c)$ with the m_α distinct power products in the non-constant indeterminates; r vanishes *identically* iff every coefficient $p_\alpha(c)$ vanishes, so $\mathcal{V} = V(p_1, \ldots, p_N)$.

Decomposing the combined system computes instead the **joint solution variety**

$$\mathrm{Sol}(A) \cap \mathrm{Sol}(P), \tag{2}$$

an *existential* condition: there *exists* a function satisfying both A and P. Projecting (2) onto the constants gives

$$\pi_c\big(\,\mathrm{Sol}(A) \cap \mathrm{Sol}(P)\big) \;=\; \mathcal{V} \;\cup\; \{\text{partial-solution strata}\} \;\supseteq\; \mathcal{V}. \tag{3}$$

Definition 1 (partial-solution stratum) *A* partial-solution stratum *is a set of constant values c at which a* proper, nontrivial *sub-family of the ansatz solutions satisfies P, while the family as a whole does not. Equivalently: $c \notin \mathcal{V}$, yet* $\mathrm{Sol}(A(c)) \cap \mathrm{Sol}(P)$ *contains more than the zero solution.*

A prerequisite is immediate from Definition 1: the ansatz solution space at c must be at least two-dimensional. A one-dimensional family either solves P in its entirety or only through its zero member; there is no proper nontrivial sub-family to peel off.

2 A two-dimensional ansatz where the strata appear

Example 1 *Take a two-dimensional ansatz and a first-order PDE,*

$$A(c): \quad \Psi'' - c\,\Psi = 0, \qquad\qquad P: \quad \Psi' - \Psi = 0. \tag{4}$$

For $c = \omega^2$ the ansatz solution space is two-dimensional, $\Psi = K_1 e^{\omega x} + K_2 e^{-\omega x}$.

Membership (core method): reduce P modulo A. The leader of P is Ψ', which is ranked *below* the leader Ψ'' of A, so P is already fully reduced with respect to A; the remainder is P itself,

$$r \;=\; \Psi' - \Psi \;=\; (+1)\cdot\Psi' + (-1)\cdot\Psi.$$

The projection coefficients are the constants $+1$ and -1: never zero, with no c-dependence. Hence

$$\mathcal{V} \;=\; V(1,-1) \;=\; \varnothing.$$

This is correct: no value of c makes the entire two-parameter family $\{K_1 e^{\omega x} + K_2 e^{-\omega x}\}$ obey the first-order condition $\Psi' = \Psi$.

Combined decomposition: adjoin P, rank the constant lowest. The decomposition selects the lower-leader equation $\Psi' - \Psi$ into the triangular set first, then reduces $\Psi'' - c\,\Psi$ against it. Using $\Psi' = \Psi \Rightarrow \Psi'' = \Psi$,

$$\Psi'' - c\,\Psi \;\longrightarrow\; \Psi - c\,\Psi \;=\; (1-c)\,\Psi \;=\; 0.$$

This equation has leader Ψ and initial $(1-c)$, so the algorithm splits on the vanishing of $(1-c)$:

cell	reduced system	solutions
$c \neq 1$	$(1-c)\Psi = 0, \quad \Psi' = \Psi$	$\Psi \equiv 0$ (trivial only)
$c = 1$	$0 = 0, \quad \Psi' = \Psi$	$\Psi = Ke^{x}$ (one-parameter)

The cell $c = 1$ is the partial-solution stratum. At $c = 1$ the ansatz has the two modes e^{x} and e^{-x}; the PDE $\Psi' = \Psi$ retains the e^{x} mode and discards e^{-x}. Exactly half the family solves the PDE — "some, but not all" — so $c = 1 \notin \mathcal{V}$ even though it is a genuine, nonempty solution stratum of the joint system.

3 Matching the orders does not help

Example 1 pairs a *second*-order ansatz with a *first*-order PDE, which invites the hope that a partial stratum is an order-mismatch artifact: raise the PDE to the ansatz's order and the gap might close. It does not. The reason is structural. Ritt's full reduction of P modulo A rewrites $\Psi^{(n)}$ (and every higher derivative) through the ansatz's order-n ODE, so the *remainder is always confined to order* $\leq n-1$, whatever the order of P. The PDE's content at and above the ansatz leader is washed out; only the reduced remainder — an operator of order $\leq n-1$ — can distinguish the ansatz modes, and whether it shares a *proper* right factor with the ansatz operator is a condition on the constants, indifferent to how P's order compares with n. What closes the gap is $n \leq 1$ (a one-dimensional solution space; §5), a property of the *ansatz*, not of the PDE or of the two orders' relation.

Example 2 (both components second order)

$$A(c): \quad \Psi'' - c\,\Psi = 0, \qquad\qquad P: \quad \Psi'' - 3\Psi' + 2\Psi = 0. \tag{5}$$

Both are second order. The ansatz has the two-mode family $\{e^{\sqrt{c}\,x}, e^{-\sqrt{c}\,x}\}$; the PDE has solutions $\{e^{x}, e^{2x}\}$.

Membership. Reducing P modulo A ($\Psi'' \equiv c\,\Psi$, with multiplier $h = 1$, so $h \neq 0$ *everywhere* — the strata below are genuine, not bad-locus artifacts) gives

$$r \;=\; c\,\Psi - 3\Psi' + 2\Psi \;=\; -3\,\Psi' + (c+2)\,\Psi.$$

The projection coefficients are -3 and $c+2$; the first never vanishes, so $\mathcal{V} = \varnothing$.

Existence. A mode $e^{\rho x}$ ($\rho = \pm\sqrt{c}$) solves P iff $\rho^2 - 3\rho + 2 = 0$, i.e. $\rho \in \{1, 2\}$, i.e. $c = \rho^2 \in \{1, 4\}$:

c	ansatz modes	solves P	other mode
1	$e^{\pm x}$	e^{x} $(1-3+2=0)$	e^{-x}: $1+3+2=6\neq 0$
4	$e^{\pm 2x}$	e^{2x} $(4-6+2=0)$	e^{-2x}: $4+6+2=12\neq 0$

So $V_{\exists\setminus\Psi} = \{1, 4\}$ while $\mathcal{V} = \varnothing$: at each of $c = 1, 4$ exactly one of the two ansatz modes survives P, a genuine partial stratum with $h \neq 0$. The GCRD test agrees: the reduced remainder is $R \sim \partial - \frac{c+2}{3}$, which right-divides $\partial^2 - c$ iff $\left(\frac{c+2}{3}\right)^2 = c$, i.e. $c^2 - 5c + 4 = (c-1)(c-4) = 0$, the order-1 (partial-stratum) case of the trichotomy. The example is Example 1 with its first-order PDE replaced by a second-order one that reduces to the same kind of order-1 remainder; the two behave identically precisely because reduction erases P's second-order part.

4 The correct selection criterion

Equation (3) warns that projecting the *whole* combined variety over-counts $\mathcal{V}$. One might hope to recover $\mathcal{V}$ by keeping the cell in which the adjoined PDE-equation collapsed to $0 = 0$; Example 1 shows this is wrong, as that cell is $c = 1$ whereas $\mathcal{V} = \varnothing$. The collapse cell is the *existential* locus, not the membership locus.

The correct criterion is **dimension preservation**: $c \in \mathcal{V}$ iff the combined solution space at c still equals the *full* ansatz solution space, i.e. the PDE removed nothing. In Example 1 no cell retains the full two-dimensional space (the $c = 1$ cell keeps only a one-dimensional slice), so $\mathcal{V} = \varnothing$, consistent with the direct reduction.

5 Contrast: a one-dimensional ansatz has no partial strata

Shrink the ansatz to one dimension and test a second-order PDE:

$$A(c): \quad \Psi' - c\,\Psi = 0 \quad (\Psi = Ke^{cx}), \qquad\qquad P: \quad \Psi'' - \Psi = 0.$$

Now P's leader Ψ'' dominates A's leader Ψ'. Reducing P modulo A gives $\Psi'' \to c^2\Psi$, hence $r = (c^2 - 1)\,\Psi$ and

$$\mathcal{V} \;=\; V(c^2 - 1) \;=\; \{+1, -1\}.$$

The combined decomposition produces the cells

$$\{\, c^2 \neq 1 : \Psi \equiv 0 \,\}, \qquad \{\, c = 1 : \Psi = Ke^{x} \,\}, \qquad \{\, c = -1 : \Psi = Ke^{-x} \,\},$$

and the nontrivial cells $c = \pm 1$ now retain the *entire* (one-dimensional) family. Here the collapse cells coincide with $\mathcal{V}$ and there are no partial strata: a one-dimensional family admits no proper nontrivial sub-family.

6 Autoreducedness does not preclude the strata: a PDE example

One might conjecture a combinatorial precondition for the partial strata: that they arise only when the reduced PDE and the ansatz still *interact* under Ritt reduction. In Example 1 the remainder $r = \Psi' - \Psi$ can reduce the ansatz equation $\Psi'' - c\,\Psi$ (its leader Ψ' has Ψ'' as a proper derivative), and it was exactly that rewriting which produced the split on $1 - c$. What if reduction terminates completely — r can reduce nothing in A and A nothing in r, so that $A \cup \{r\}$ is autoreduced? Does that preclude a partial-solution stratum?

In the ordinary-differential setting (one derivation, one unknown) the conjecture holds, but only vacuously. There A is a single equation with leader $\Psi^{(k)}$, and the reduced remainder involves only $\Psi, \ldots, \Psi^{(k-1)}$; if r involves the unknown at all, its leader $\Psi^{(j)}$ has $\Psi^{(k)}$ as a proper derivative, so r *can* always reduce A. The hypothesis therefore forces $r = p(c)$, a pure constants condition — and then both computations return $V(p)$ carrying the full ansatz family: dimension-preserving, no strata.

With several derivations, leaders can be incomparable and the conjecture fails.

Example 3 *Two derivations ∂_x, ∂_y:*

$$A(c): \quad \Psi_{yy} - c\,\Psi = 0, \qquad\qquad P: \quad \Psi_x - \Psi = 0.$$

P is already reduced with respect to A (Ψ_x is not a derivative of Ψ_{yy}), so $r = P$; and r can reduce nothing in A, whose equation contains only Ψ_{yy} and Ψ, neither a derivative of Ψ_x. Thus $A \cup \{r\}$ is autoreduced — and even *coherent*: the leaders' least common derivative is Ψ_{xyy}, and the Δ-polynomial

$$\partial_y^2\, r - \partial_x\, A \;=\; (\Psi_{xyy} - \Psi_{yy}) - (\Psi_{xyy} - c\,\Psi_x) \;=\; c\,\Psi_x - \Psi_{yy} \;\longrightarrow\; c\,\Psi - c\,\Psi \;=\; 0.$$

The combined decomposition therefore emits $A \cup \{r\}$ unchanged, as a single cell, with no splitting at all.

Membership: the projection coefficients of r are $+1$ and -1, so $\mathcal{V} = \varnothing$ — correct, since the general ansatz solution $\Psi = F(x)\,e^{\omega y} + G(x)\,e^{-\omega y}$ ($\omega^2 = c$; F, G arbitrary functions) does not satisfy $\Psi_x = \Psi$.

Joint variety: $\Psi_x = \Psi$ forces $\Psi = e^x h(y)$, and then $h'' = c\,h$, so

$$\Psi \;=\; e^x\big(K_1 e^{\omega y} + K_2 e^{-\omega y}\big)$$

— nontrivial at *every* c, and a proper subfamily of the ansatz family (the arbitrary functions F, G collapse to constant multiples of e^x). Every c is a partial-solution stratum; here the stratum is the whole parameter space.

The structural reading: "$A \cup \{r\}$ is autoreduced and coherent" means the decomposition has no more algebra to do — it is precisely a finished cell — *not* that r imposes no constraint. The cell's solution set is the joint variety, and whenever that sits strictly between 0 and $\mathrm{Sol}(A(c))$ at some $c \notin \mathcal{V}$, a partial stratum is present. What precludes the strata is r having no non-constant part, $r = p(c)$ — which, in the single-derivation case, is the only way the autoreducedness hypothesis can be met at all. The strata are a fact of the solution geometry, not of the rewriting combinatorics.

7 Reading

The membership formulation (1) asks: *for which ODE-coefficients c is the PDE an automatic consequence of the ansatz?* — so that the ansatz family is, as a whole, a PDE-solution family. The partial strata answer a different question: *for which c does the ansatz family happen to contain a PDE solution it was not designed to?* — coincidental solutions living on a proper sub-family at special parameter values. They are real solutions, but invisible to the membership formulation by construction.

For the hydrogen computation this means: adjoining the Schrödinger equation to the ansatz and decomposing surfaces parameter values where a degenerate, lower-dimensional slice of the ansatz solves the PDE while the generic member does not. If the goal is the membership locus $\mathcal{V}$, either reduce P modulo A directly, or — having decomposed the combined system — keep only the dimension-preserving cells. Projecting the full joint variety answers the existential question and returns $\mathcal{V}$ together with the partial strata.